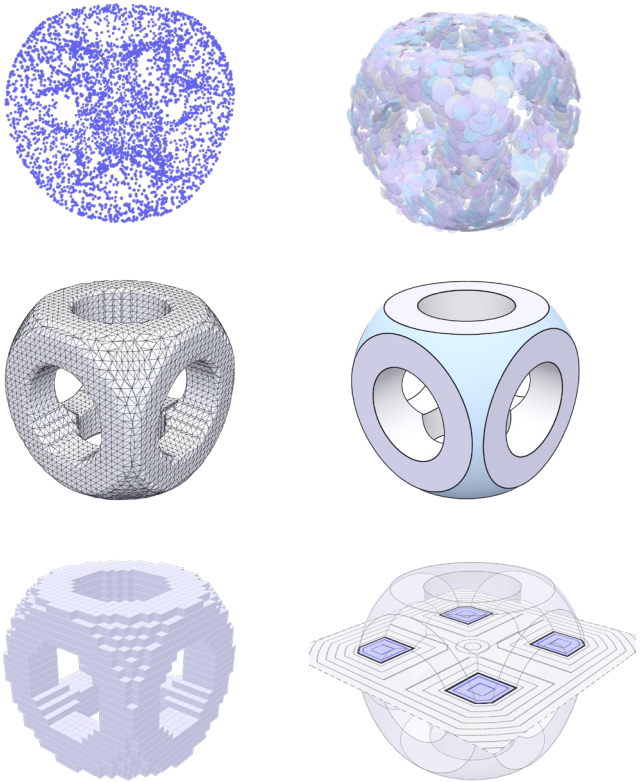
Computational Geometry

Lecture 2.0 | 26 October 2026

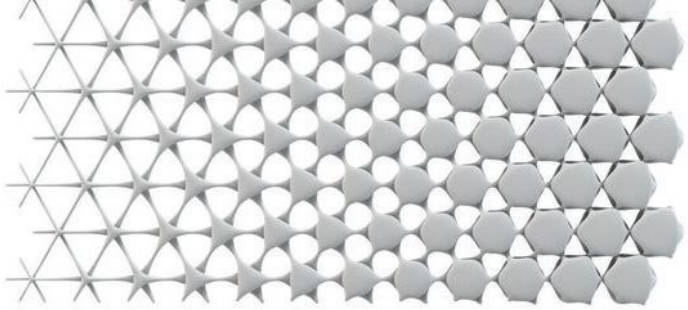


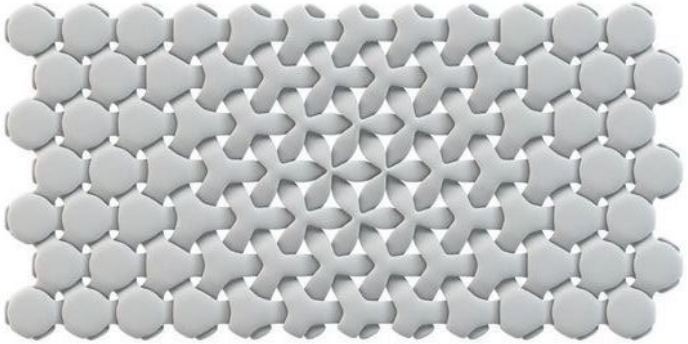
Learning Goals

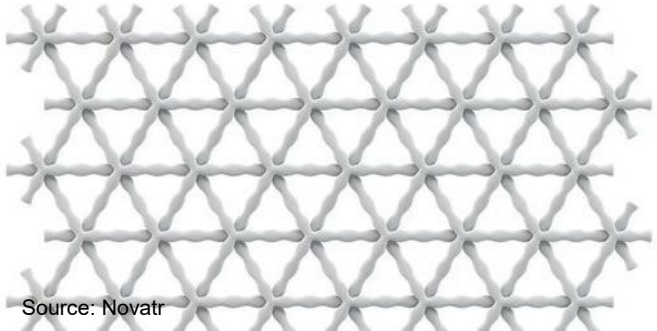
1. Gain an overview of the **history** of Computational Design in Architectural design and Fabrication.

2. Understand **Computational Geometry** in 2D and 3D and how to construct them parametrically.

3. Understand the **Operations** on Computational Geometry.





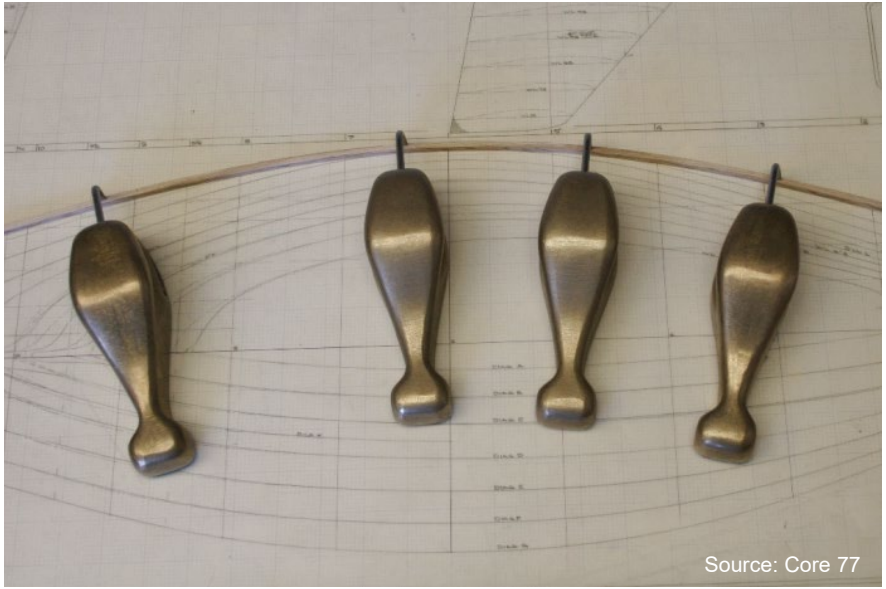


Source: Novatr

Geometry Before the Computer



Drafting table



Spline: Thin wooden strip bent by weights (“ducks”)

Brief History of Computational Design in Architecture


Source: Ivan Sutherland

SketchPad at MIT
(Ivan Sutherland)

1963

Computer-aided-design

Brief History of Computational Design in Architecture

SketchPad at MIT (Ivan Sutherland)

In-house CAD Tools at SOM Computer Group

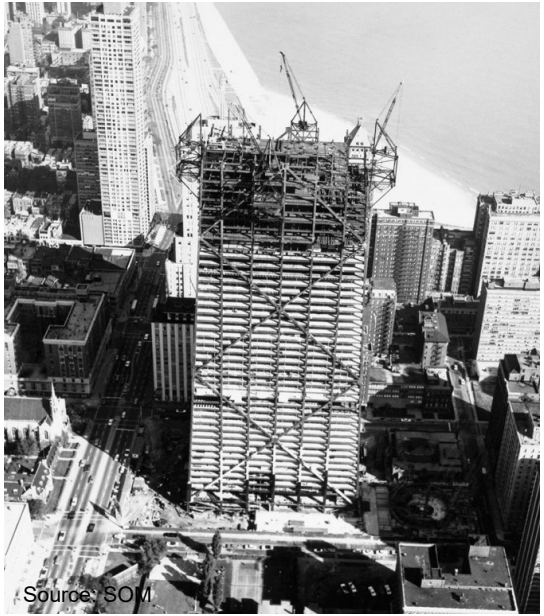
1963

1970s

Computer-aided-design →



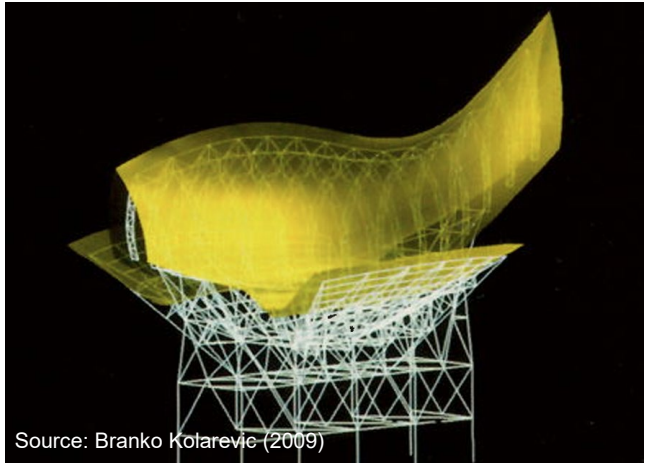
Sears Tower (1971)



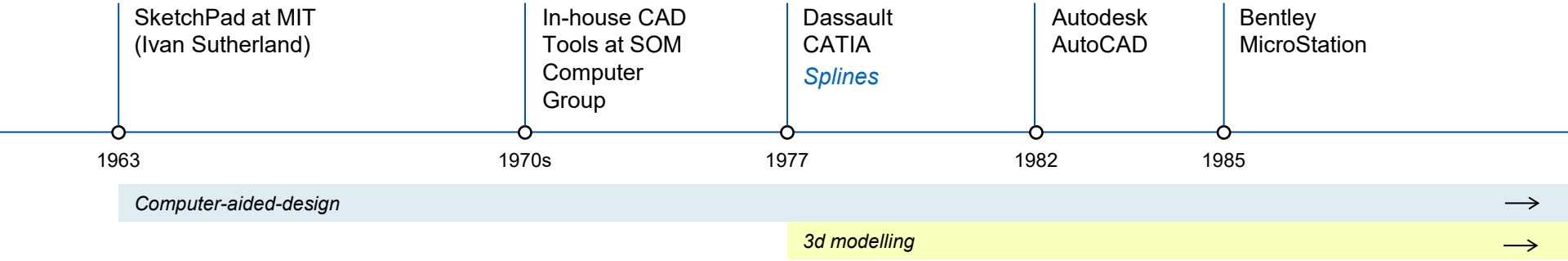
John Hancock Center

Brief History of Computational Design in Architecture

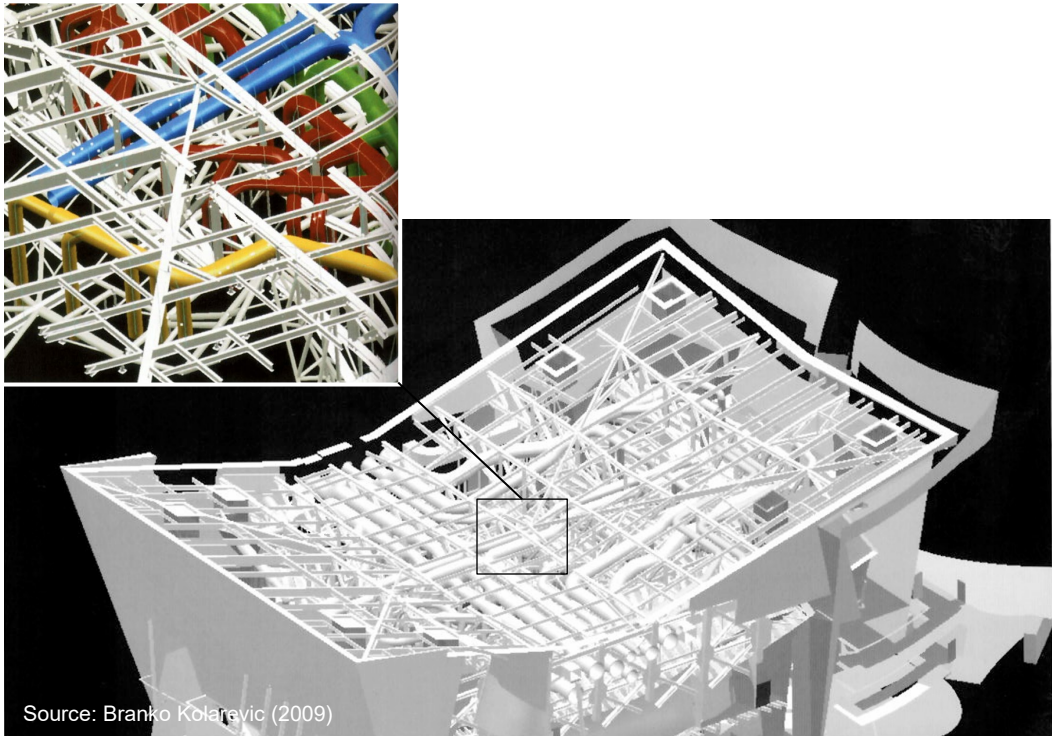




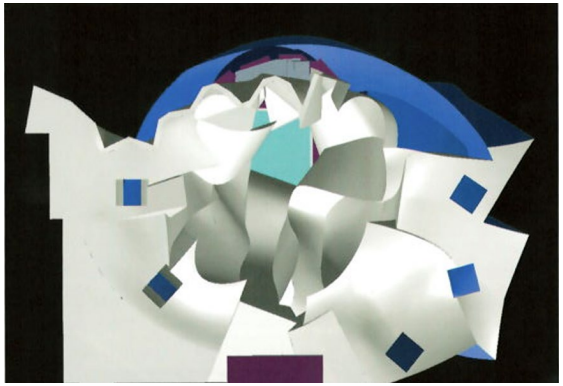
Golden Fish Sculpture for 1992 Olympics Barcelona by Frank Gehry Modelled in CATIA

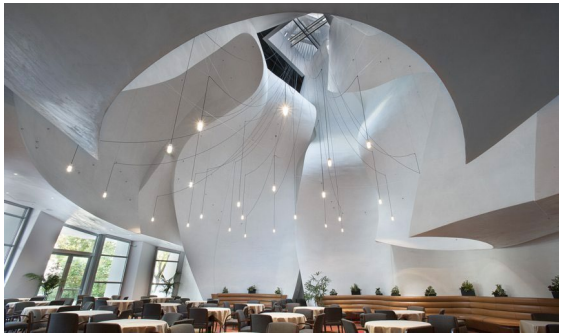


Brief History of Computational Design in Architecture



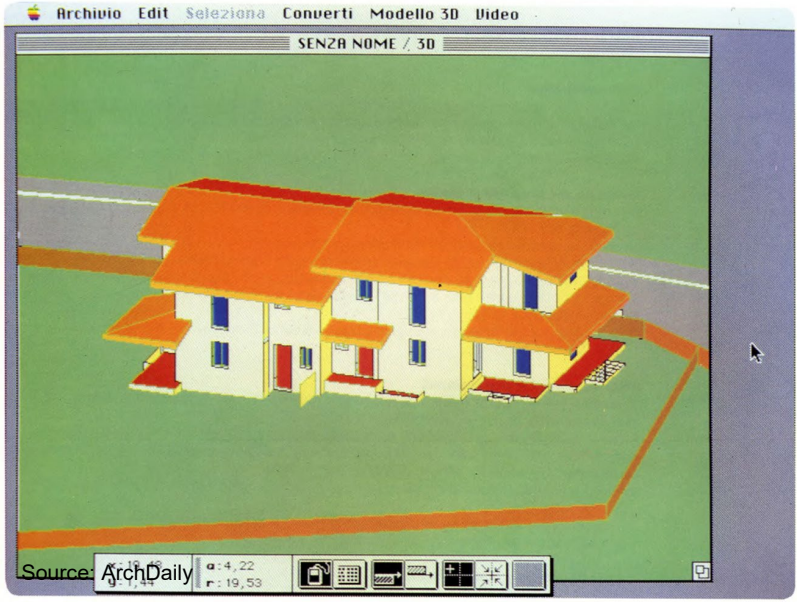
Composite model of Walt Disney Concert Hall, Frank Gehry (1993-03)
Modelled using CATIA





Modelling of Founders' room interior
Modelled using CATIA

Brief History of Computational Design in Architecture



Screenshot from Radar CH (later ArchiCAD) 1984

ArchiCAD
BIM

Dassault CATIA — 1977

Autodesk AutoCAD — 1982

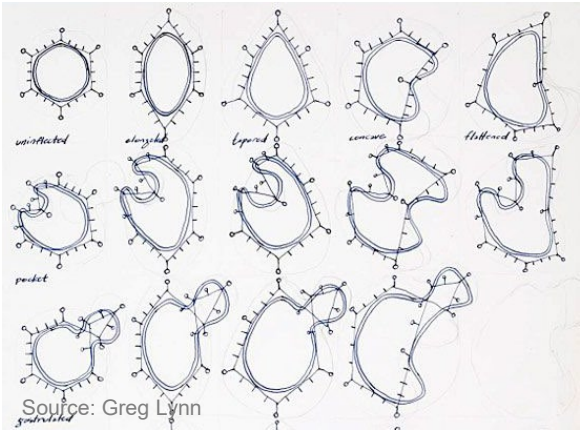
Bentley MicroStation — 1985

1987

← *Computer-aided-design* →

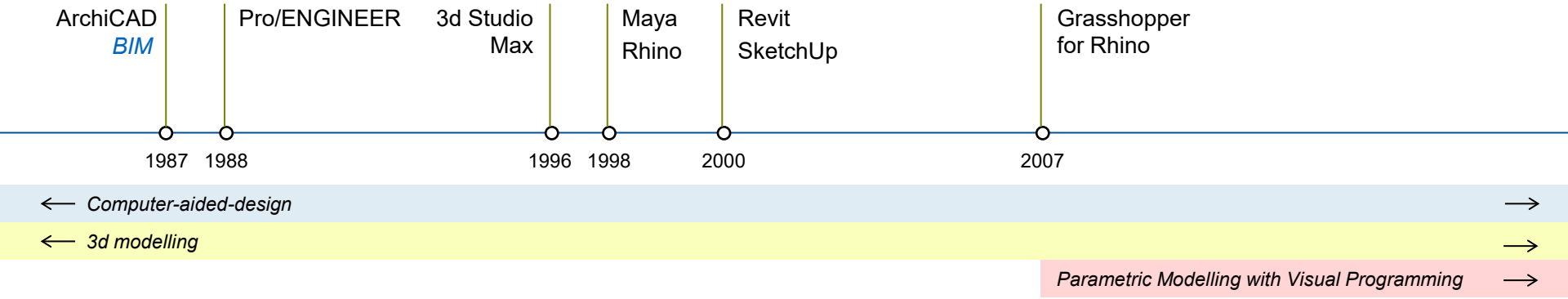
3d modelling →

Brief History of Computational Design in Architecture

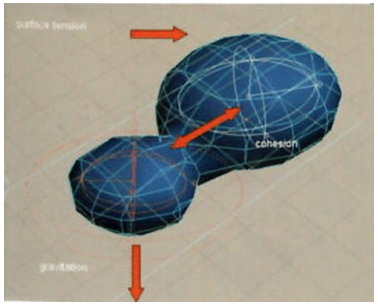


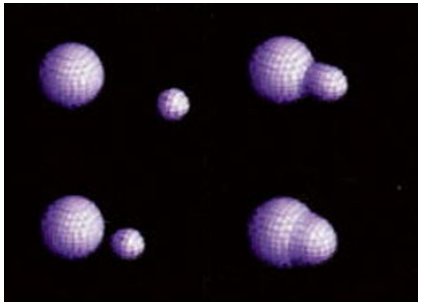


Parametric design of "Embryological House" by Greg Lynn (1997-2001)



Brief History of Computational Design in Architecture









Bubble pavilion for BMW,1999 (“Blobs” or “Metaballs”)





Digital Fabrication of Foam moulds and Aluminium elements

Contents

1

Fundamentals

- Coordinate system
- Point
- Vector
- Plane

2

2D Geometry

- Line Segments
- Circles and Arcs
- Polylines
- Freeform curves

3

3D Geometry

- Solid Primitives
- Boolean Operators
- Freeform NURBS surfaces
- B-Reps
- Meshes

4

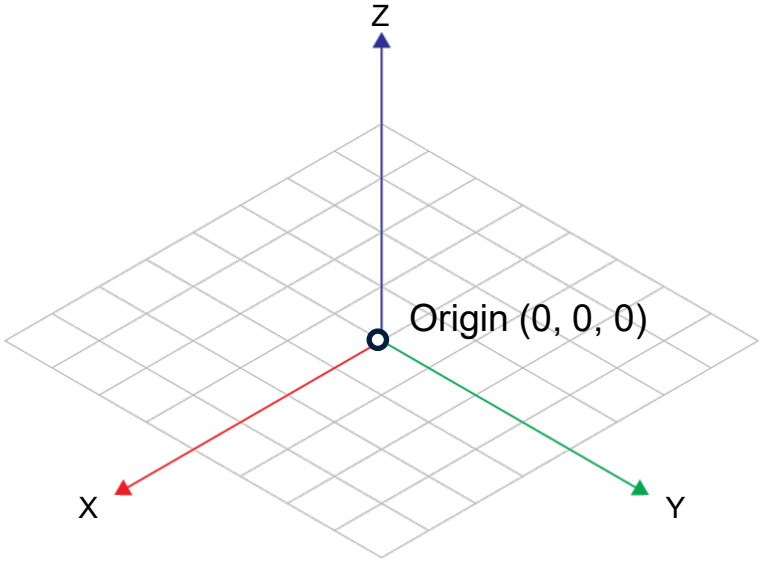
Transformations

1 /

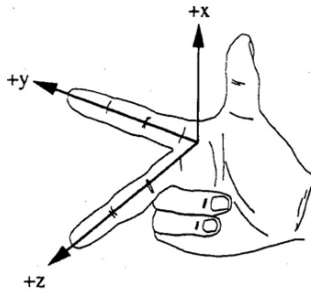
Geometry fundamentals

1. Coordinate systems
2. Point
3. Vector
4. Plane

Global Coordinate System

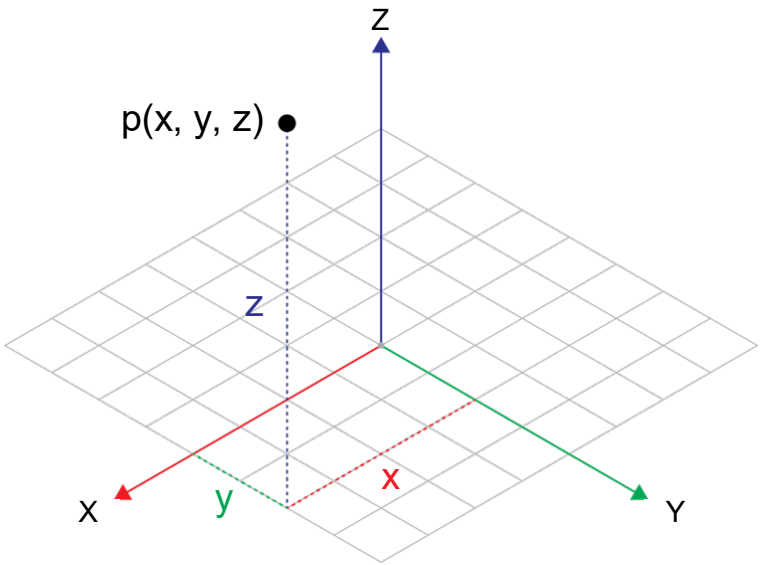


Cartesian Coordinate System



Right-hand rule

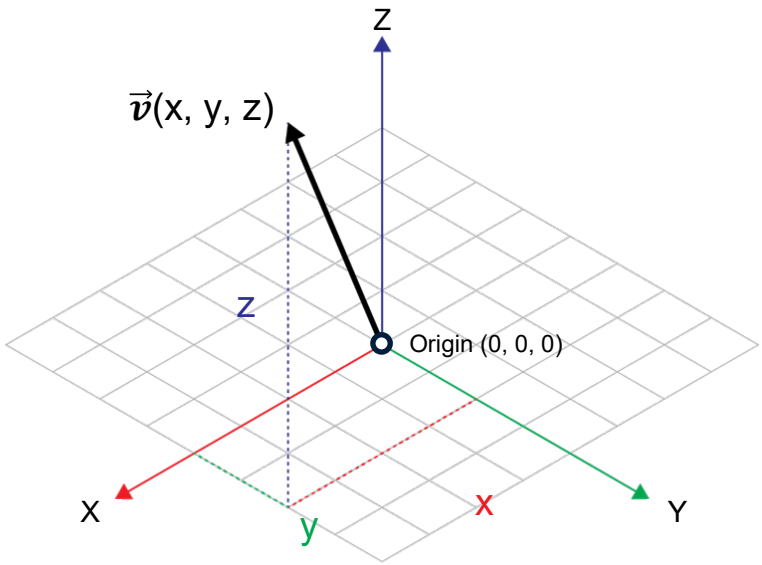
Point



WITH COORDINATES

> float *x*
> float *y*
> float *z*

Vector



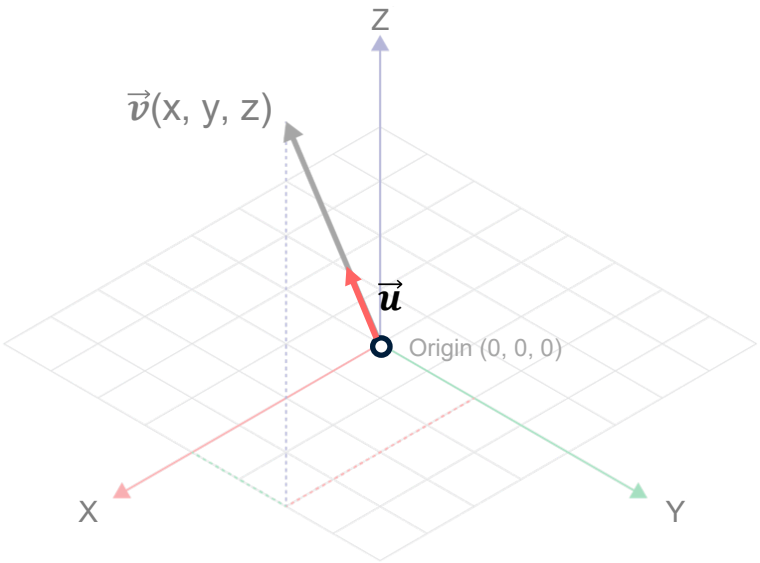
WITH X, Y, Z COMPONENTS

- float *x*
- float *y*
- float *z*

PROPERTIES

- magnitude $|| \vec{v} ||$

Unit Vector



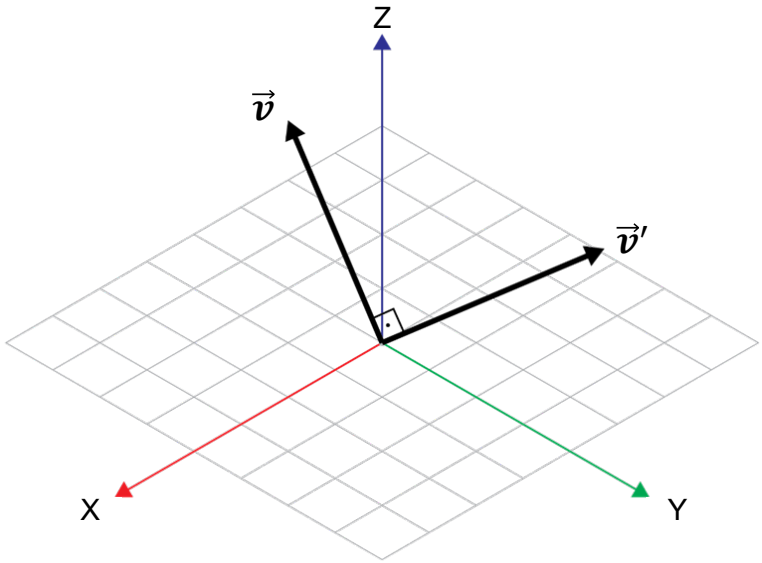
WITH X, Y, Z COMPONENTS

- float *x*
- float *y*
- float *z*

PROPERTIES

- magnitude $||\vec{u}|| = 1$

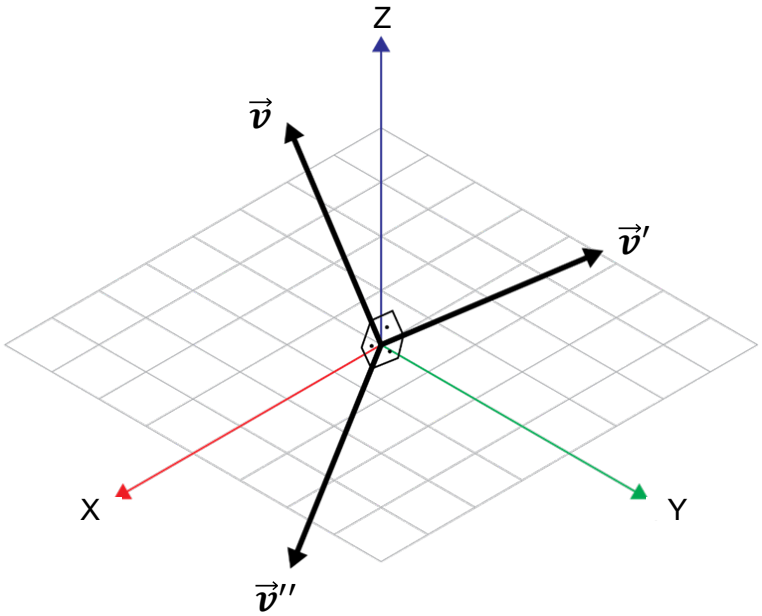
Orthogonal Vectors



PROPERTIES

angle $\angle(\vec{v}, \vec{v}') = 90^0$

Orthonormal Vectors

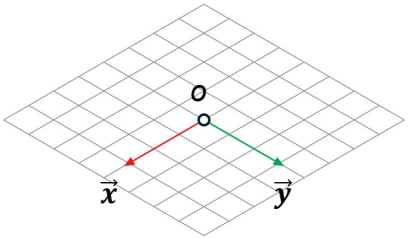


PROPERTIES

angle $\angle(\vec{v}, \vec{v}') = 90^0$
angle $\angle(\vec{v}', \vec{v}'') = 90^0$
angle $\angle(\vec{v}'', \vec{v}) = 90^0$

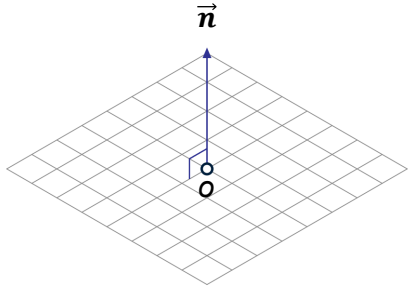
X-axis
Y-axis
Z-axis
are orthonormal vectors

Plane



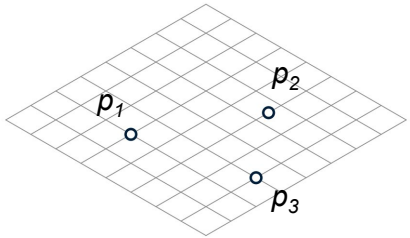
WITH POINT, X, Y AXES

- point *o* (float x_o, float y_o, float z_o)
- *vector* $\vec{x}$ (float x_x, float y_x, float z_x)
- *vector* $\vec{y}$ (float x_y, float y_y, float z_y)



WITH POINT, NORMAL VECTOR

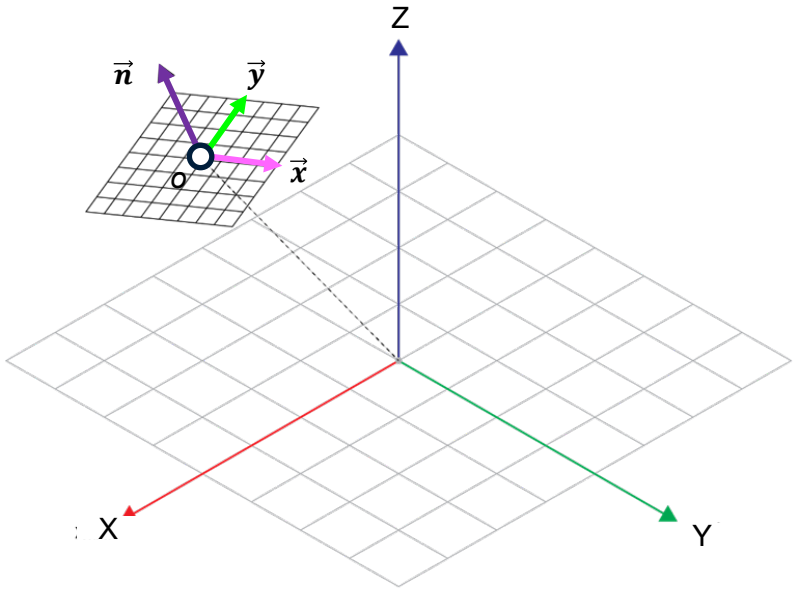
- point *o* (float x_o, float y_o, float z_o)
- *vector* $\vec{n}$ (float x_n, float y_n, float z_n)

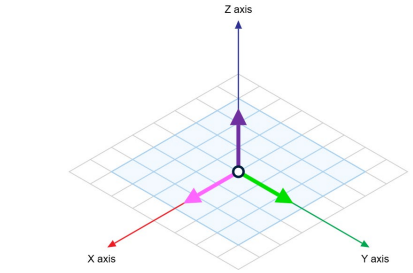


WITH THREE POINTS

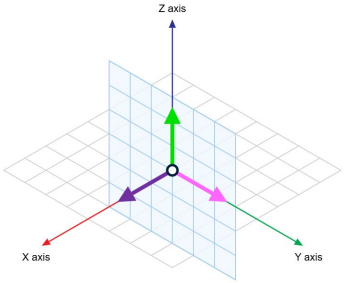
- point p_1 (float x_1, float y_1, float z_1)
- point p_2 (float x_2, float y_2, float z_2)
- point p_3 (float x_3, float y_3, float z_3)

Plane

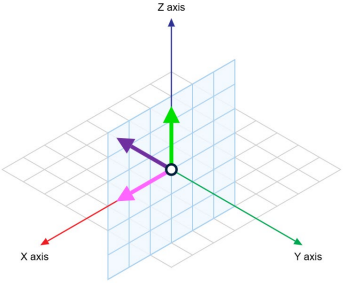




World XY plane



World YZ plane



World XZ plane

origin local X local Y local Z

PROPERTIES

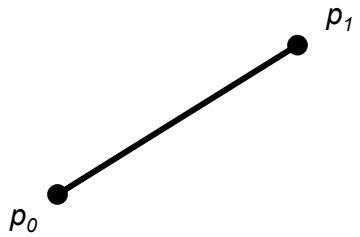
- origin o
- x-axis $\vec{x}$
- y-axis $\vec{y}$
- z-axis (normal vector) $\vec{n}$

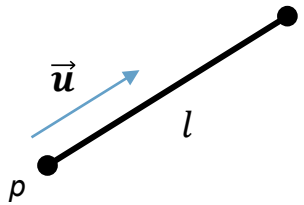
2 /

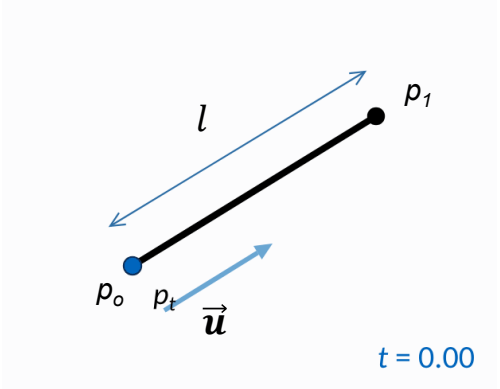
2D Geometry

1. Line Segments
2. Circles and Arcs
3. Polylines
4. Freeform Curves

Line Segments







WITH TWO POINTS

- point p_0 (float x_0, float y_0, float z_0)
- point p_1 (float x_1, float y_1, float z_1)

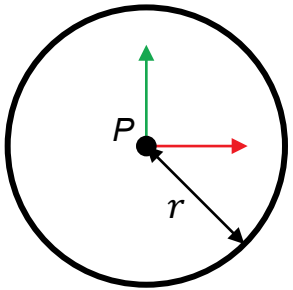
WITH POINT, UNIT VECTOR, LENGTH

- point p_0 (float x_0, float y_0, float z_0)
- *vector* $\vec{u}$ (float x_n, float y_n, float z_n)
- float l

PROPERTIES

- length l
- *direction vector* $\vec{u}$
- *end points* p_0, p_1
- *point on line* p_t *with parameter* $t \in [0, 1]$

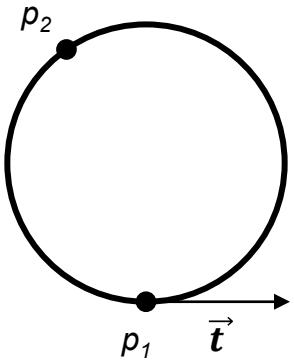
Circles



WITH POINT, X, Y AXES

plane P ((float x_o, float y_o, float z_o)
(float x_x, float y_x, float z_x)
(float x_y, float y_y, float z_y))

float r

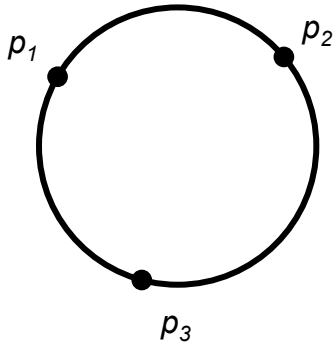


WITH TWO POINTS, TANGENT VECTOR

point p_1 (float x_1, float y_1, float z_1)

point p_2 (float x_2, float y_2, float z_2)

vector $\vec{t}$ (float x_t, float y_t, float z_t)



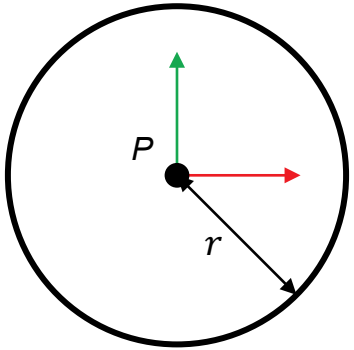
WITH THREE POINTS

point p_1 (float x_1, float y_1, float z_1)

point p_2 (float x_2, float y_2, float z_2)

point p_3 (float x_3, float y_3, float z_3)

Circle Properties

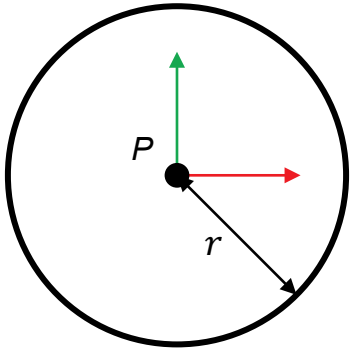


PROPERTIES

plane P ((float x_o, float y_o, float z_o)
(float x_x, float y_x, float z_x)
(float x_y, float y_y, float z_y))

radius r

Circle Properties



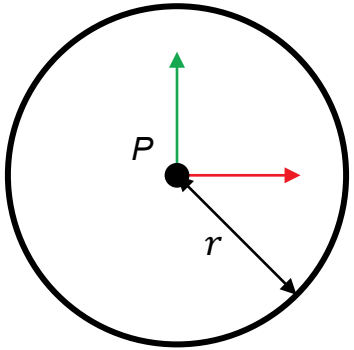
How would you get a point and tangent vector on the circle?

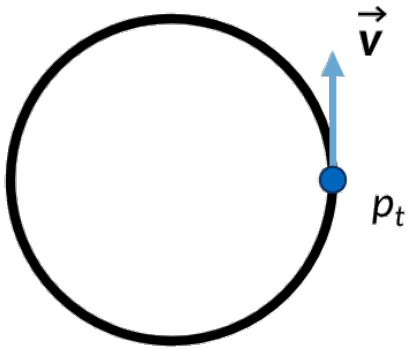
PROPERTIES

plane P ((float x_o, float y_o, float z_o)
(float x_x, float y_x, float z_x)
(float x_y, float y_y, float z_y))

radius r

Circle Properties





$t = 0.00\pi$

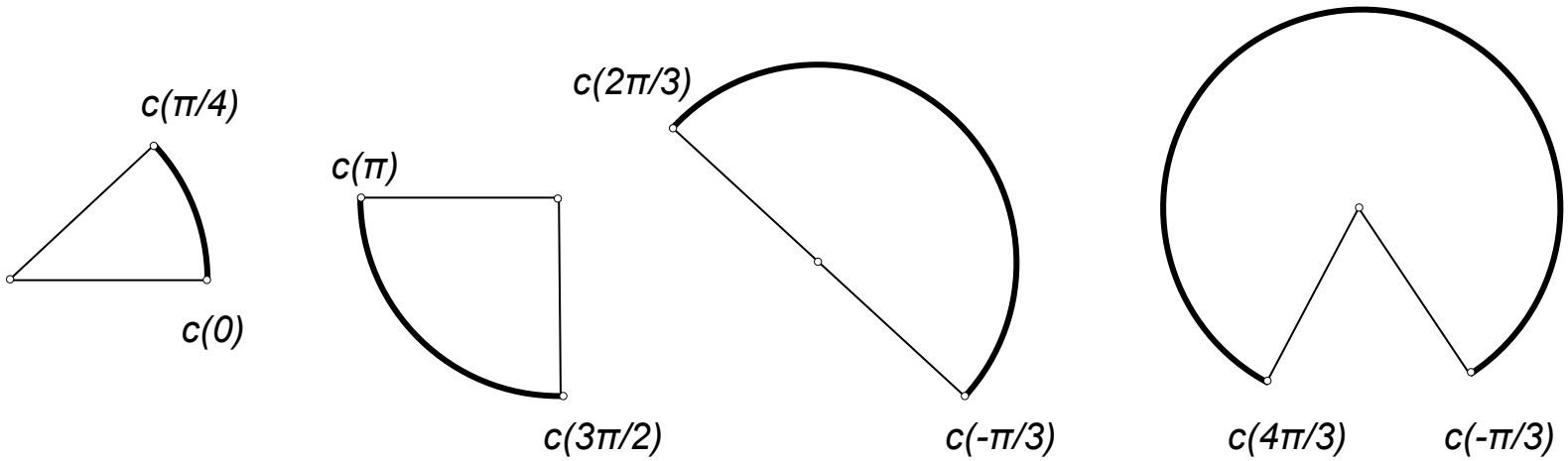
PROPERTIES

plane P ((float x_o, float y_o, float z_o)
(float x_x, float y_x, float z_x)
(float x_y, float y_y, float z_y))

radius r

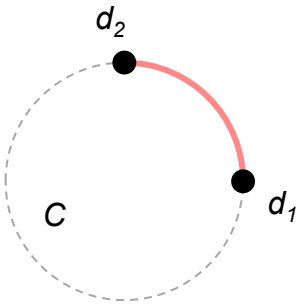
point p_t and
tangent vector $\vec{v}$
with parameter $t \in [0, 2\pi]$

Arcs



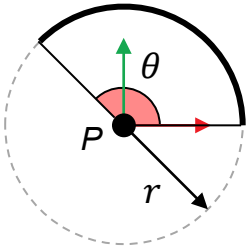
Arcs as sub-domains of circles

Arcs



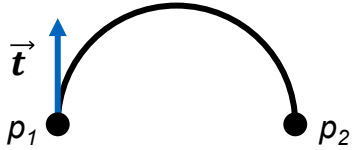
WITH CIRCLE AND DOMAIN

- circle C
- float d_1
- float d_2



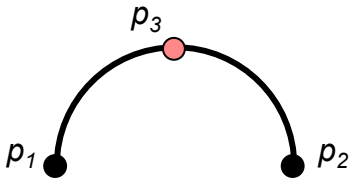
WITH PLANE, RADIUS, ANGLE

- plane P
- float r
- float θ



WITH TWO POINTS, TANGENT

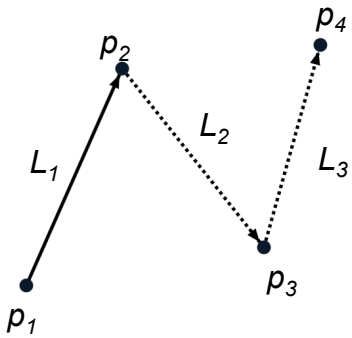
- point p_1
- point p_2
- *vector* $\vec{t}$



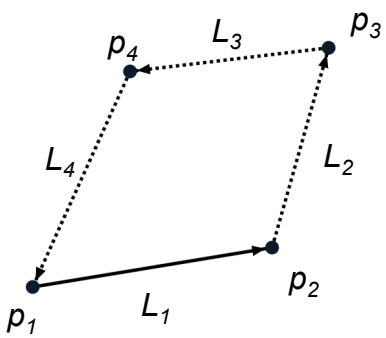
WITH THREE POINTS

- point p_1
- point p_2
- point p_3

Polylines



Open polyline



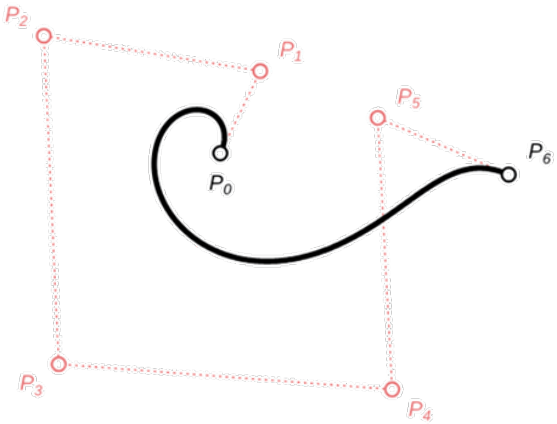
Closed polyline

A **polyline** is a sequence of straight line segments.

If the **first and last points coincide**, the polyline forms **a closed polygon.**

Freeform Curves

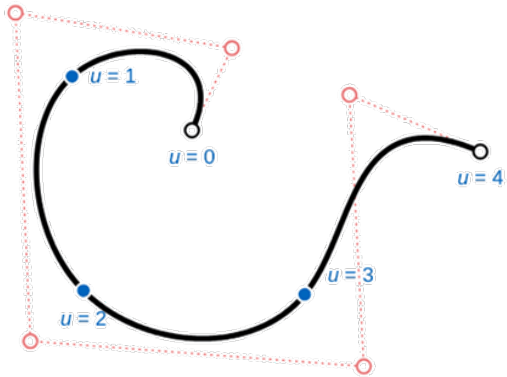
Bézier curve



control points

- no local control

B-spline curve

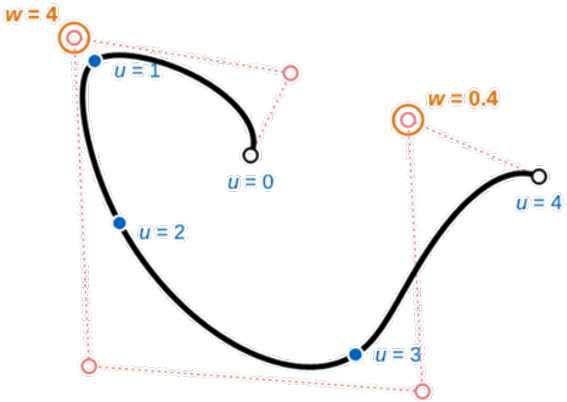


control points

knots

- Cannot create a circle

NURBS curve



control points

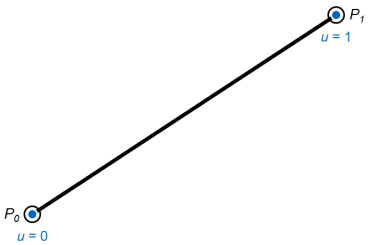
knots

weights

+ Can define all shapes

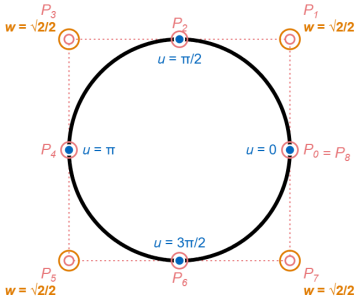
NURBS Curves

Line



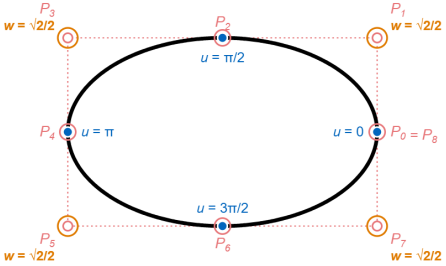
knot vector $U = \{ 0, 0, 1, 1 \}$
degree 1 · 2 control points · all weights w = 1

Circle



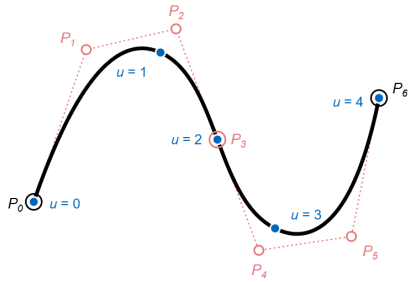
$U = \{ 0, 0, 0, \pi/2, \pi/2, \pi, \pi, 3\pi/2, 3\pi/2, 2\pi, 2\pi, 2\pi \}$
degree 2 · 9 control points (first = last) · others w = 1

Ellipse



$U = \{ 0, 0, 0, \pi/2, \pi/2, \pi, \pi, 3\pi/2, 3\pi/2, 2\pi, 2\pi, 2\pi \}$
degree 2 · 9 control points (first = last) · others w = 1

Free-form curve



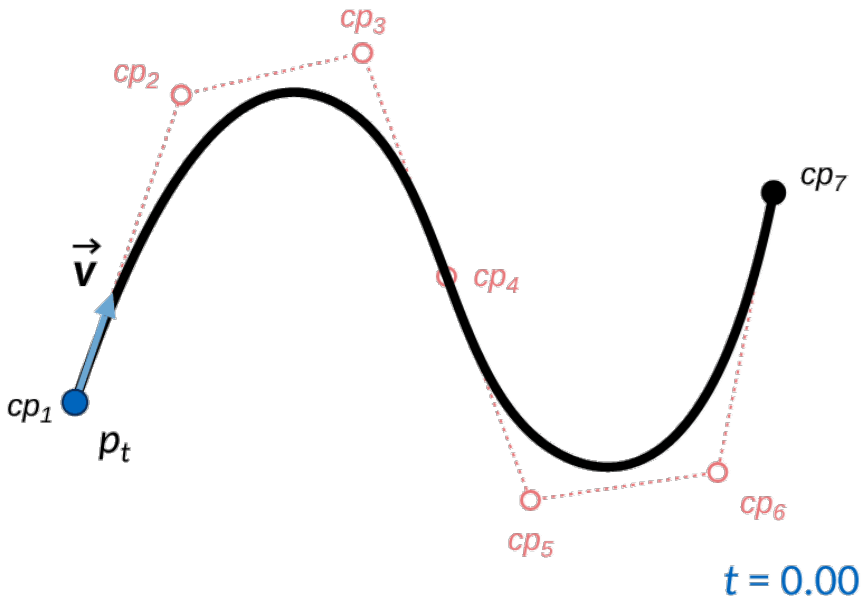
knot vector $U = \{ 0, 0, 0, 0, 1, 2, 3, 4, 4, 4, 4 \}$
degree 3 · 7 control points · all weights w = 1

NURBS Curve Properties

PROPERTIES

control points [cp1, cp2 … cpn]
knot vector [k1, k2, … kn]
weights [w1, w2, … wn]

point p_t and
tangent vector $\vec{v}$
with parameter $t \in [0, 1]$

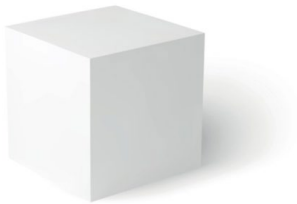


3 /

3D Geometry

1. Solid Primitives
2. Boolean Operators
3. Freeform NURBS Surfaces
4. B-Reps
5. Meshes

Solid Primitives



Cube / Box



Cone



Cylinder



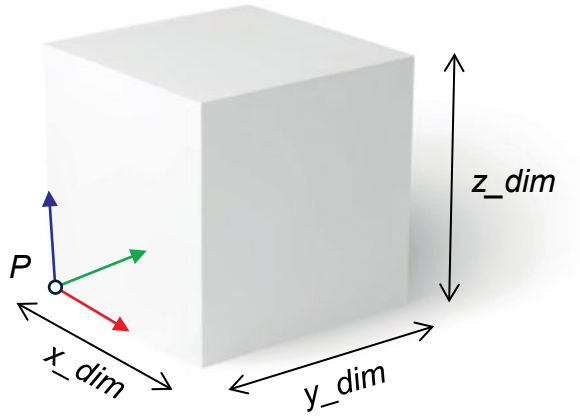
Sphere



Torus

Solid Primitives

Cube / Box

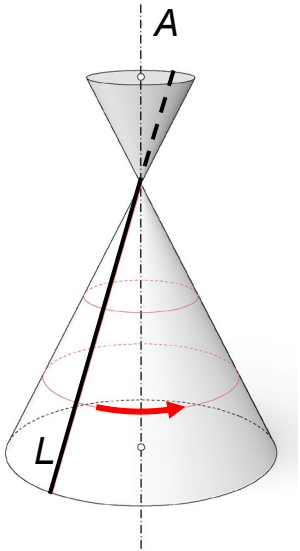


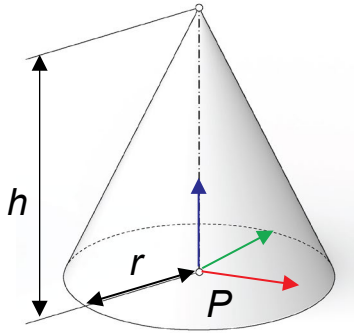
WITH PLANE AND THREE DIMENSIONS

```
plane P
float x_dim
float y_dim
float z_dim
```

Solid Primitives

Cone





WITH TWO LINES (REVOLUTION ABOUT AN AXIS)

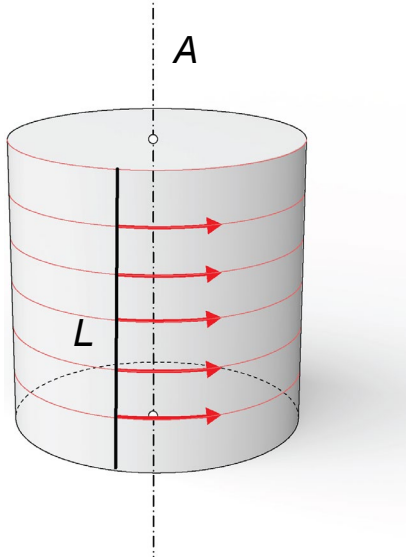
line *A*
line *L*

WITH PLANE, RADIUS, HEIGHT

plane *P*
float r
float h

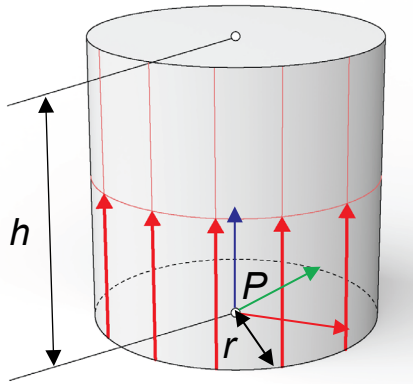
Solid Primitives

Cylinder



WITH TWO LINES (REVOLUTION ABOUT AN AXIS)

line *A*
line *L*

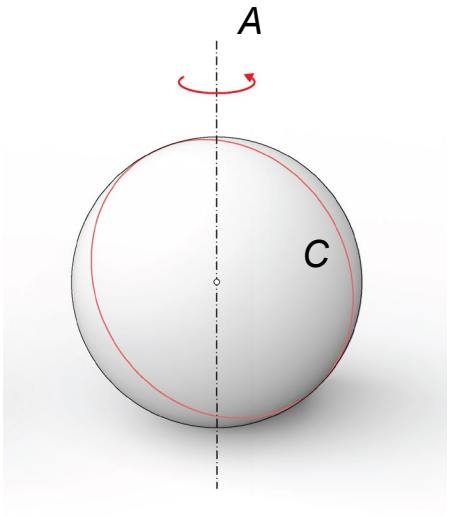


WITH PLANE, RADIUS, HEIGHT

plane *P*
float r
float h

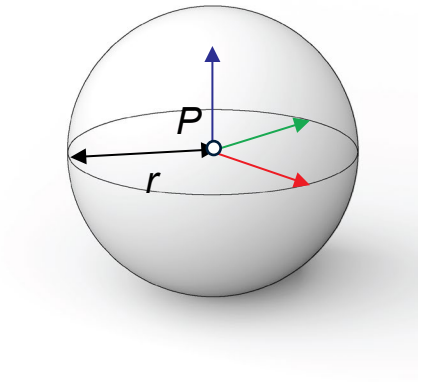
Solid Primitives

Sphere



WITH LINE & CIRCLE (REVOLUTION ABOUT AXIS)

line *A*
circle *C*

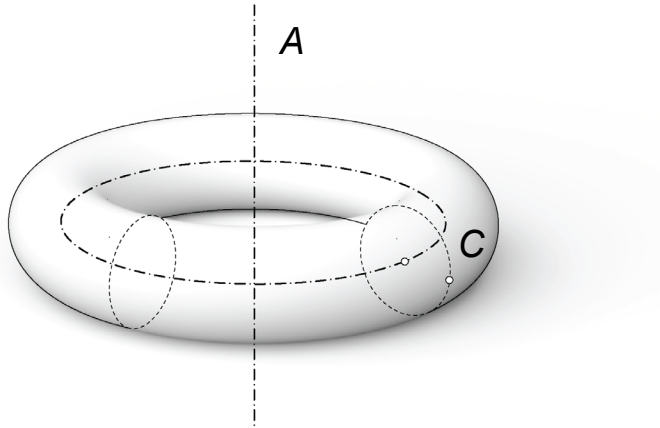


WITH PLANE, RADIUS

plane *P*
float r

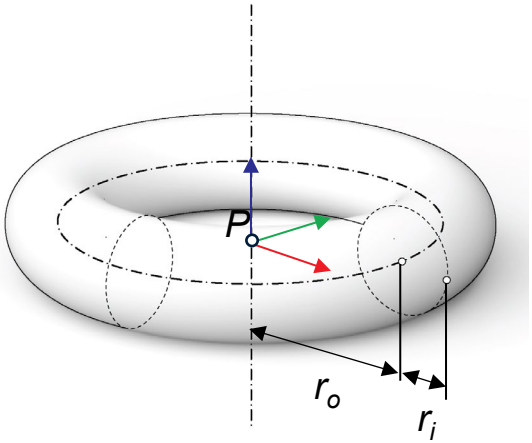
Solid Primitives

Torus



WITH LINE & CIRCLE (REVOLUTION ABOUT AXIS)

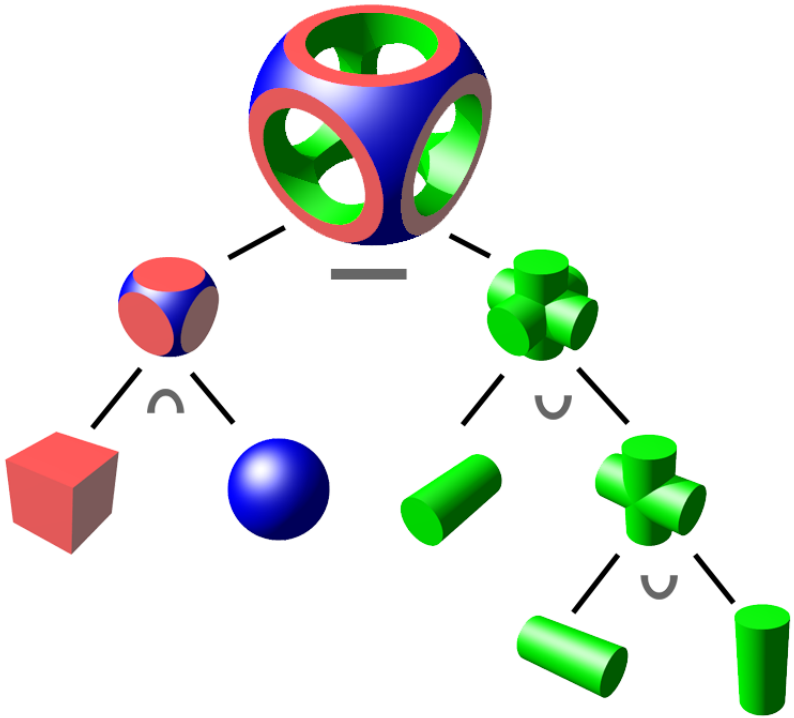
line *A*
circle *C*

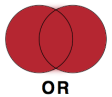


WITH PLANE, OUTER AND INNER RADII

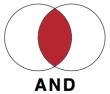
plane *P*
float r_o
float r_i

Composite Solids through Boolean Operations

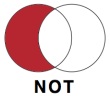




∪ : Union

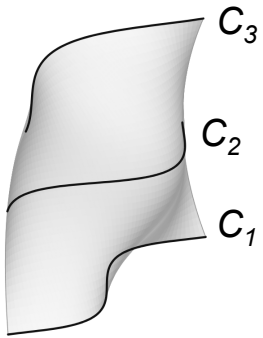


∩ : Intersection



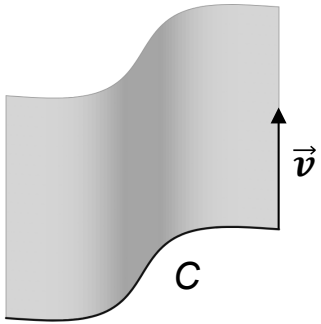
– : Difference

Freeform NURBS Surfaces



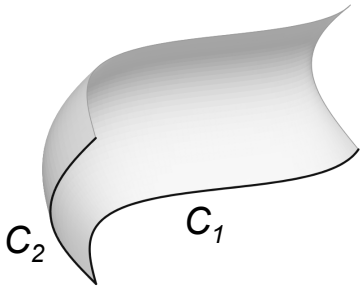
LOFT

list of curves
$[C_1, C_2 \ldots C_n]$



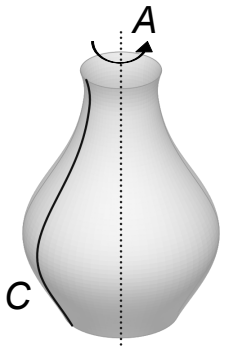
EXTRUDE

curve C
vector $\vec{v}$



SWEEP

curve C_1
curve C_2



REVOLUTION

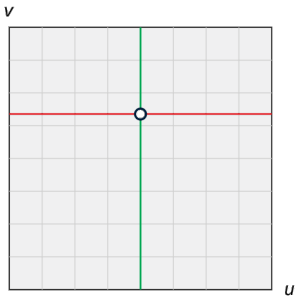
curve C
line A

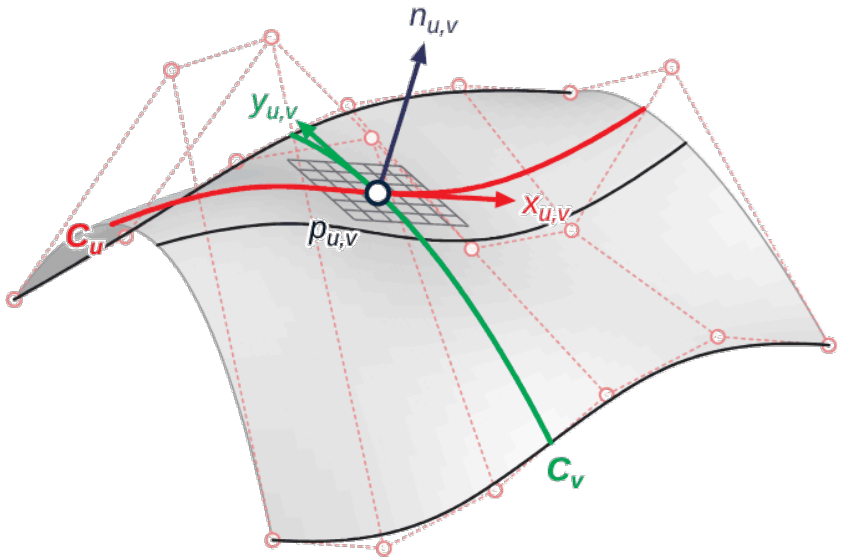
NURBS Surface Properties

PROPERTIES

control points [cp_1, cp_2 … cp_n]
control polygons [CP_1, CP_2 … CP_n]
2 knot vectors [k_{u1}, k_{u2} … k_{un}],
[k_{v1}, k_{v2} … k_{vn}],
weights [w_1, w_2 … w_n]

point $p_{u,v}$
plane $P_{u,v}$
with vector $x_{u,v}$, vector $y_{u,v}$
and normal vector $n_{u,v}$
isocurves C_u, C_v
with parameters $\boldsymbol{u, v} \in [\mathbf{0}, \mathbf{1}]$



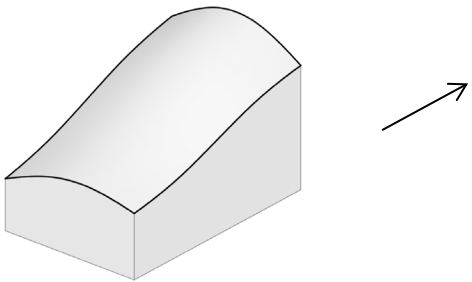


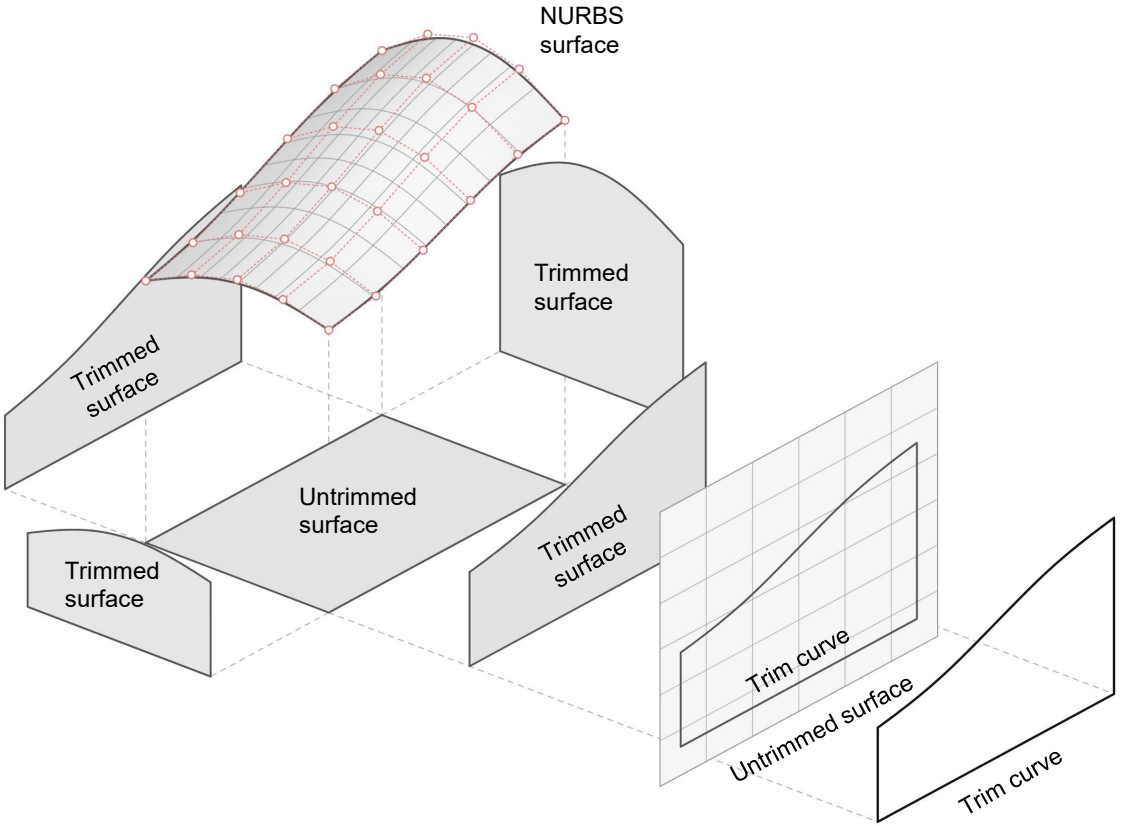
$(u, v) = (0.50, 0.67)$

Boundary Representations (B-Reps)

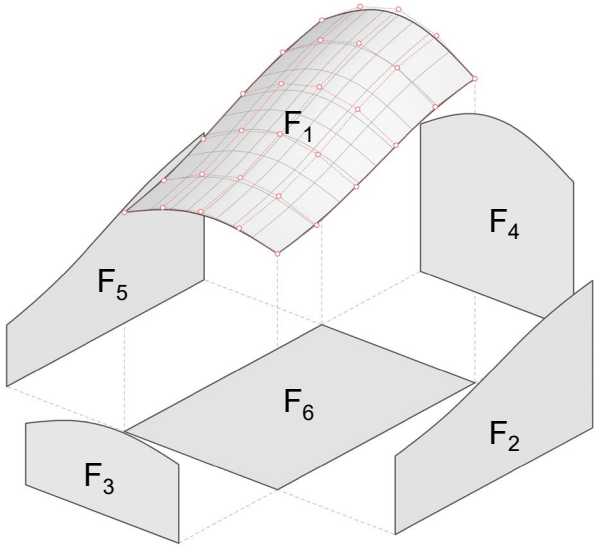
Boundary Representation is a method used in solid modelling and computer-aided design to represent the shape of a 3D object **using its geometric boundaries**.

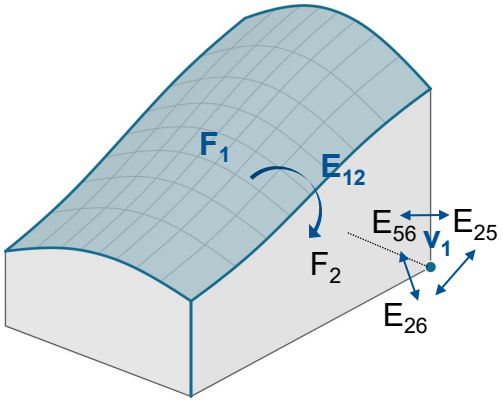
B-Reps are composed of **Faces, Edges and Vertices**





B-Rep Properties





GEOMETRY

properties of each face F

TOPOLOGY

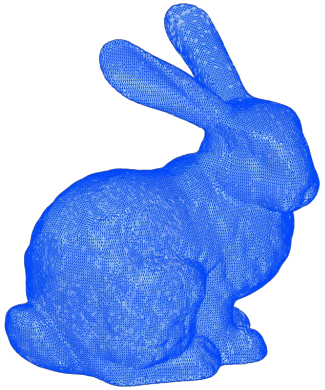
connectivity graph between
faces and edge
edges and vertices

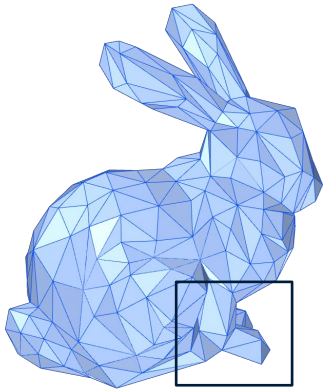
Meshes

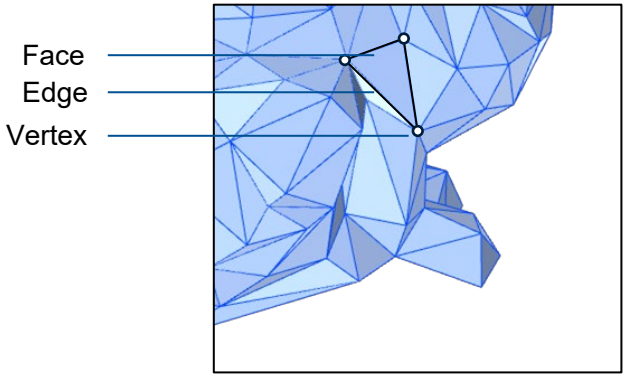
Meshing is the process of converting a continuous geometric model, typically created in CAD, into a discrete representation made up of small elements such as triangles, quadrilaterals, tetrahedra, or hexahedra.

Meshes are composed of **Faces, Edges and Vertices.**

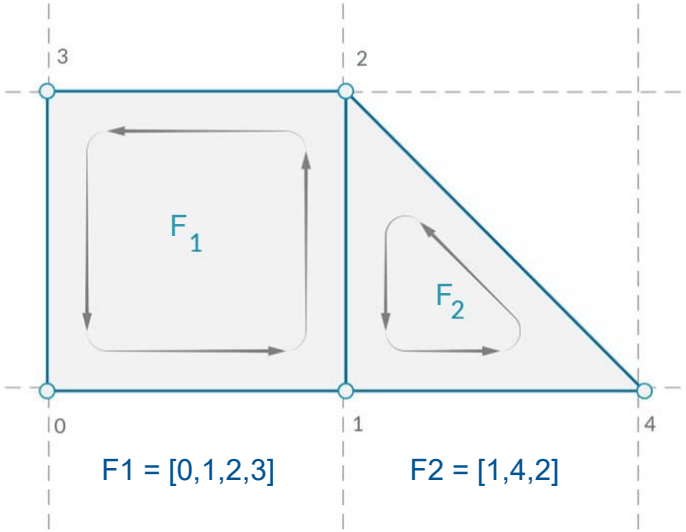
The faces could be **triangles, Quads, hexagons or n-gons.**

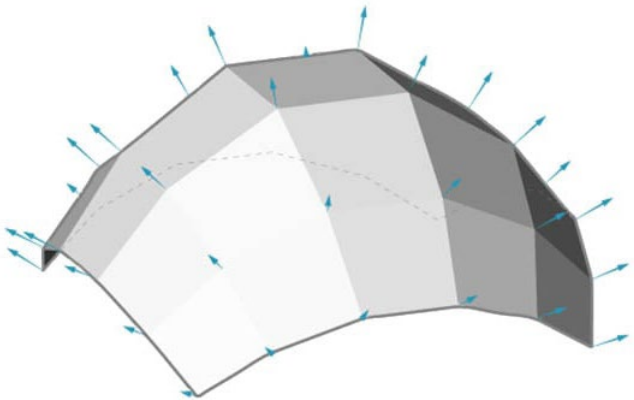






Mesh Properties





Vertex normals and face normals

GEOMETRY

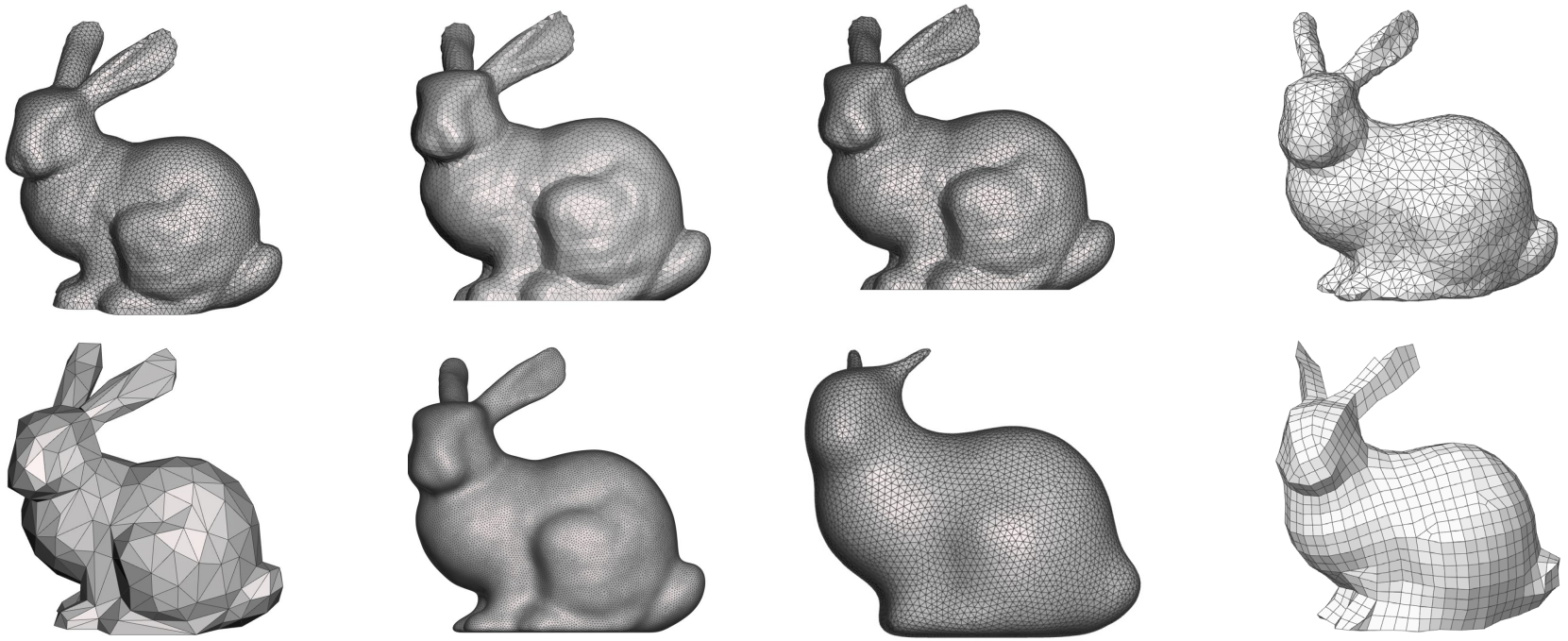
Faces: Ordered list of vertices
Edges: List of vertices, List of faces
Vertices: Coordinates
Face and vertex normal vectors

TOPOLOGY

Connectivity graph between
faces and edge
edges and vertices

Image source: Autodesk, The Dynamo Primer, “Meshes”, 2023.

Mesh Operations



Simplification **Subdivision** **Smoothening** **Remeshing**

NURBS Surfaces vs Meshes

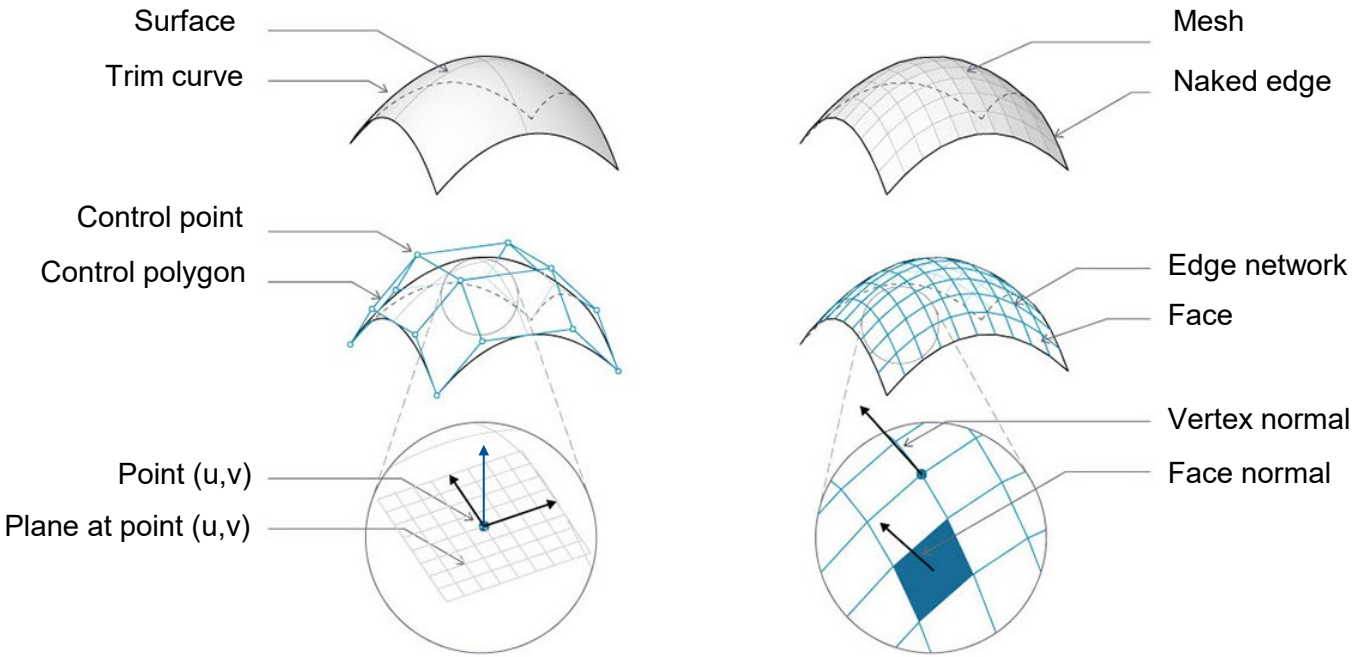
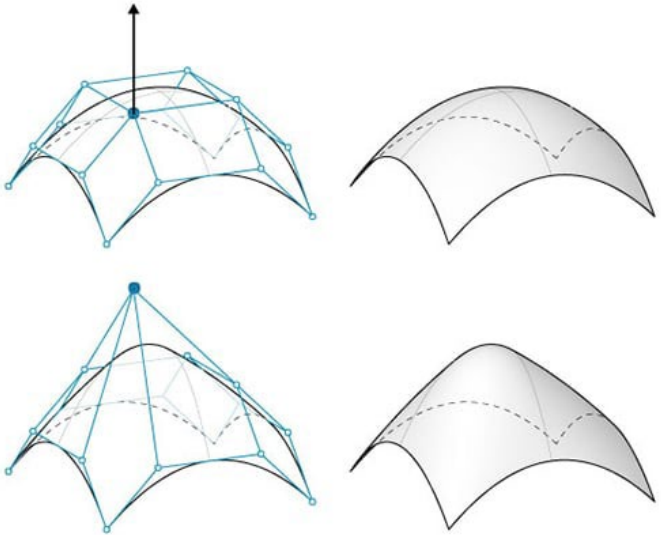
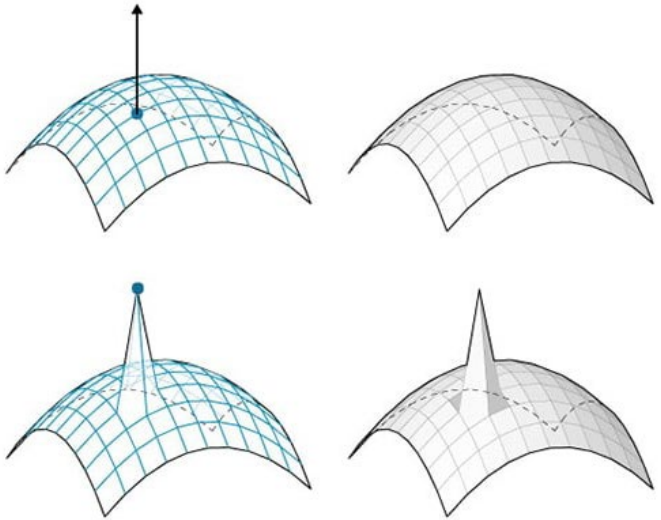


Image source: Autodesk, The Dynamo Primer, “Meshes”, 2023.

NURBS Surfaces vs Meshes



Global control



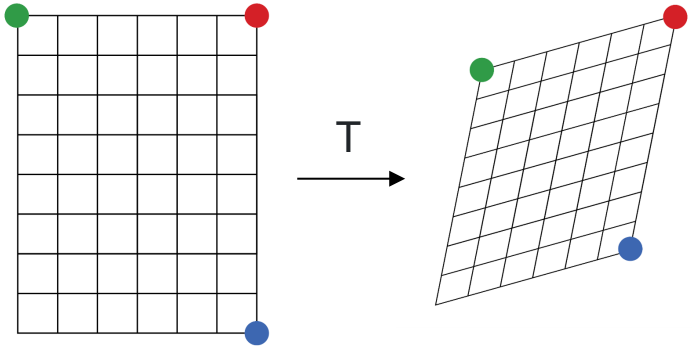
Local control

Image source: Autodesk, The Dynamo Primer, “Meshes”, 2023.

4 /

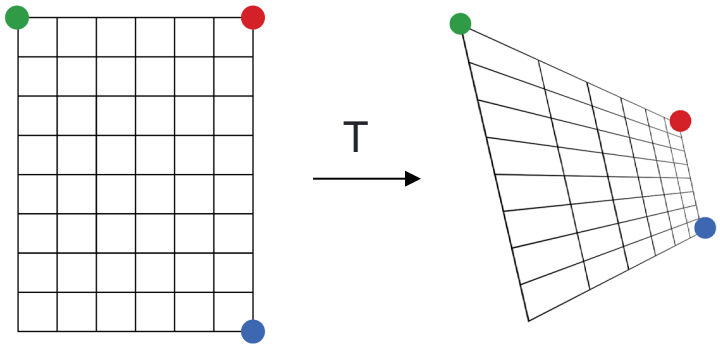
Transformations

Transformations



Affine

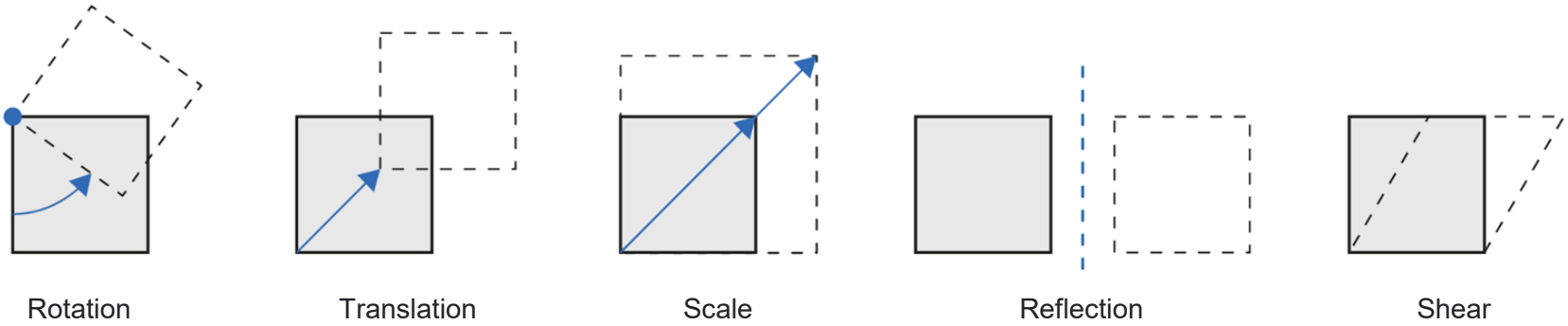
Affine transformations such as rotation and translation **preserve straight lines, parallelism**, and the ratios of distances between points



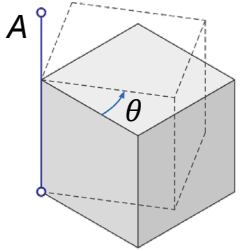
Non-affine

Non-affine transformations, such as projective or non-linear warping, **do not preserve parallelism** between lines.

Affine Transformations

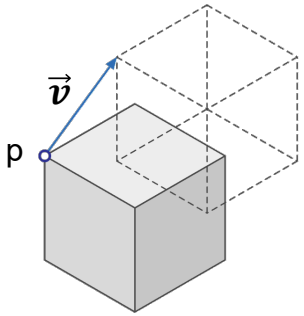


Affine Transformations



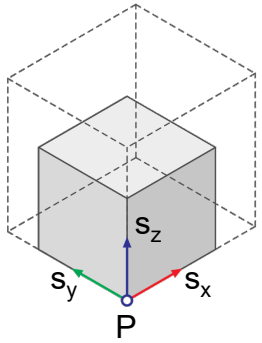
ROTATION

- line *A*
- float θ



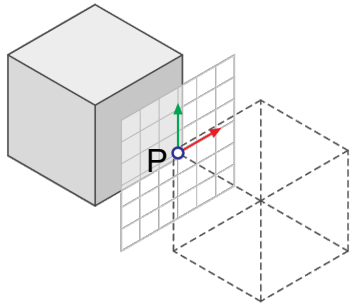
TRANSLATION

- point *p*
- vector $\vec{v}$



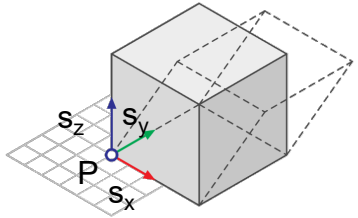
SCALE

- plane *P*
- float s_x
- float s_y
- float s_z



REFLECTION

- plane *P*



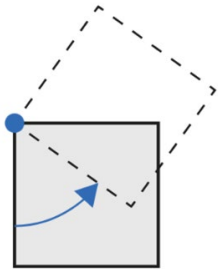
SHEAR

- plane *P*
- float s_x
- float s_y
- float s_z

Affine Transformation Matrices

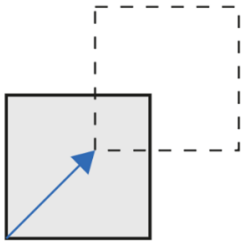
Rotation

$$T = \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 \\ a_{21} & a_{22} & a_{23} & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



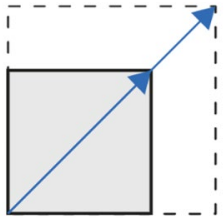
Translation

$$T = \begin{bmatrix} 1 & 0 & 0 & a_{14} \\ 0 & 1 & 0 & a_{24} \\ 0 & 0 & 1 & a_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Scale

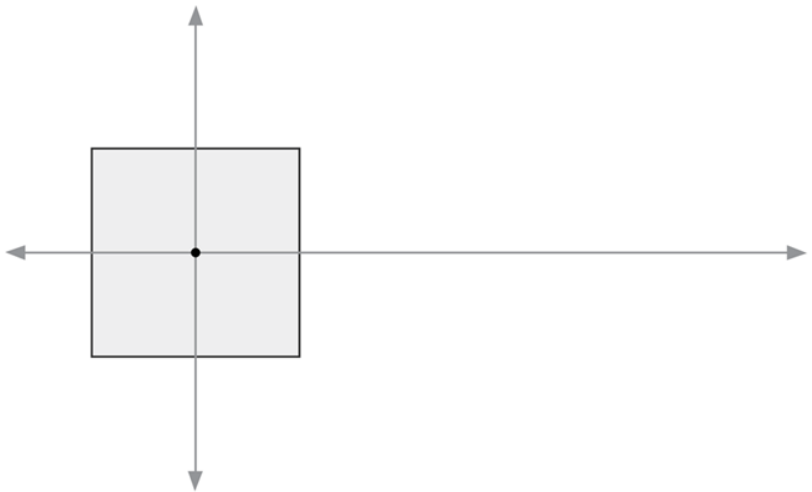
$$T = \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ 0 & a_{22} & 0 & 0 \\ 0 & 0 & a_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



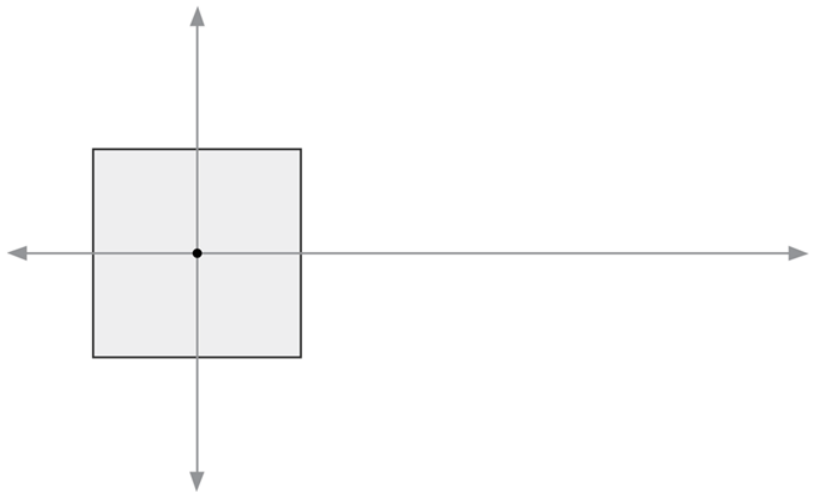
Composition

$$\underbrace{\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Rotation } \mathbf{A}} \times \underbrace{\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Translation } \mathbf{B}} = \underbrace{\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Composition } \mathbf{C = AB}}$$

Noncommutativity $(\mathbf{A} \times \mathbf{B} \neq \mathbf{B} \times \mathbf{A})$

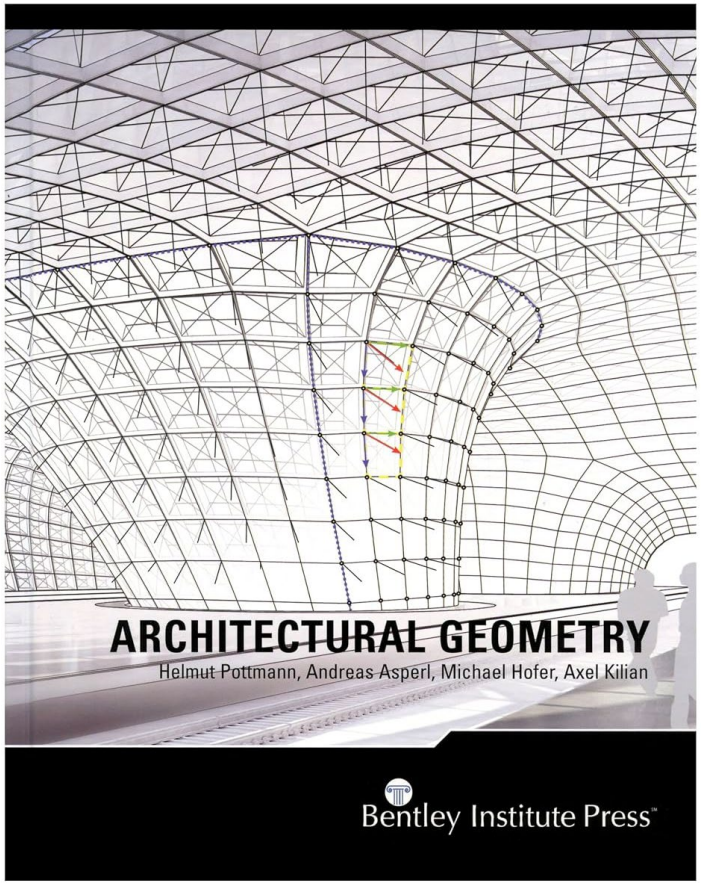


$\mathbf{C} = \mathbf{A} \times \mathbf{B}$ (apply $\mathbf{B}$ to the right of $\mathbf{A}$)



$\mathbf{C} = \mathbf{B} \times \mathbf{A}$ (apply $\mathbf{B}$ to the left of $\mathbf{A}$)

Reading Material



Pottmann, H., Asperl, A., Hofer, M., Kilian, A.: **Architectural Geometry**. Bentley Institute Press, Exton, PA, 2007.